

01/06/26 "Kamini Dhana Sri Keerthi"

Working on: "Presentation"

Title "[Tentative]":

Comparative study on Pinhole Defected
as TSV in terms of Electrical
characterisation

① Comparative [with pinhole defect & without
pinhole defect]

② Studying how a pinhole defect affects
the electrical behaviour of as TSV

a₁ Electrical behaviour parameters
such as S parameters & performance
parameters.

a₂ S parameters [scattering parameters]

definition: Parameters used to measure how
signals travel through a circuit especi-
ally at high frequencies.

- S_{11} [Input Reflection Coefficient]
- S_{21} [Forward transmission is g
coefficient]

S_{11} [Input Reflection Coefficient]

definition: How much of the input signal
bounces back.

S_{21} [forward TC]

definition: How much signal reaches
the o/p [measuring].

slide 02:

Via first; Via middle; via last

or Via first:

1.1 TSV process

[holes-mark-drill-fill]

1.2 In via first TSV process is done

first and then the other fabrication
parts follow. (before CMOS
transistor fabrication).

CMOS fabrication.

± Complementary metal oxide semicon,
± definition: CMOS transistor fabrication
is a process of creating NMOS &
PMOS on a silicon wafer to build
an IC [Integrated circuit]

↳ chip contains billions of RC
Transistors

FEOL (Front end of line)

± it is a process of transistor fabrication

BEOL (Back end of line)

FEOL - Making Transistors

BEOL - Making Metal connections
between transistors

Via first

Via fab₁ → FEOL + BEOL → Polish

Via Middle

FEOL → Via fab₁ → BEOL → Po

Via last

FEOL → BEOL → Polish

→ Via fab₁

2D ⇒ Transistor - Metal wire → Trans₁

3D ⇒ Trans₁
↓
TSV
↓
Trans

slide 03

Fabrication process

→ Refer day 05 written notes.

* wafer thinning

1 ↓ thickness of chip

2 Both side accessible TSV.

16th equation

surrounding TSV

G_{leak} : Insulation property

$G_{leak} \uparrow$ leakage $\uparrow$

$G_{leak} \downarrow$ Better insulation.

$$G_{leak} = \frac{2\pi \epsilon_{SiO_2} (\lambda_{TSV} - 4\pi) f \epsilon_0 \epsilon_{air}}{\ln \left(\frac{r_{SV} + t_{ox}}{r_{TSV}} \right)}$$

17th equation

$$G_{leak} = \frac{2\pi \epsilon_0 r_{ox} (\lambda_{TSV} - j r_{TSV})}{\ln \left(\frac{r_{SV} + t_{ox}}{r_{TSV}} \right)}$$

Why Pinhole defect.

most common & frequent defect we will find in insulating material that effects Electrical performance by $\uparrow$ leakage & degrading signal integrity

→ Version - 2 Pdf.

Correcting PDF

✓ References → add 2 more.

2 Parameter → diagram in tab

3 Via slide

1 add Tem

2 Read why we select what via process.